
LiDAR Crime Scenes for VR

Project Proposal
COMP3018 - Professional Experience
Western Sydney University

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Executive Summary

The LiDAR Crime Scenes for VR project is designed to transition a proven mixed reality application from the now unsupported Microsoft HoloLens 2 to the Meta Quest 3 platform. Leveraging the Quest 3's passthrough and depth sensing technologies, this project aims to deliver a scalable, cost-effective solution tailored to the needs of forensic investigations.

In Collaboration with our industry partners, The Mobile Apps Man and ForensicXR, Unity will be utilised in developing the software. The application will enable users to select standard 3D scene files from the user device and render them into an interactive virtual environment for both individual and collaborative sessions. This approach not only expands the application's capabilities by harnessing the advanced features of the Meta Quest 3, but also provides a cost-effective solution that meets the evolving needs of modern forensic investigations.

Introduction

This proposal is developed in collaboration with our industry partner, The Mobile Apps Man and ForensicXR, led by Kirk Duncan. Our partner is dedicated to advancing innovative mixed reality solutions for forensic investigations.

Last year, a team developed a holographic crime scene application for the Microsoft HoloLens 2, enabling multiple users to participate in a shared investigative session. However, with the HoloLens platform no longer getting further support, there is a need for transitioning to a new hardware. Building on this proof-of-concept, the current project aims to extend support to Meta Quest 3. By leveraging the Quest 3's advanced passthrough and depth sensing capabilities.

Using Unity as the tool in development, the project will allow users to select 3D scene files stored on the device. These files are then rendered into interactive virtual scenes that users can walk through, manipulate, and scale during both individual and collaborative sessions. This new approach seeks not only to extend the application's capabilities to harness the advanced features of the Meta Quest 3, but also to deliver a scalable, cost-effective solution tailored to the evolving needs of forensic investigations.

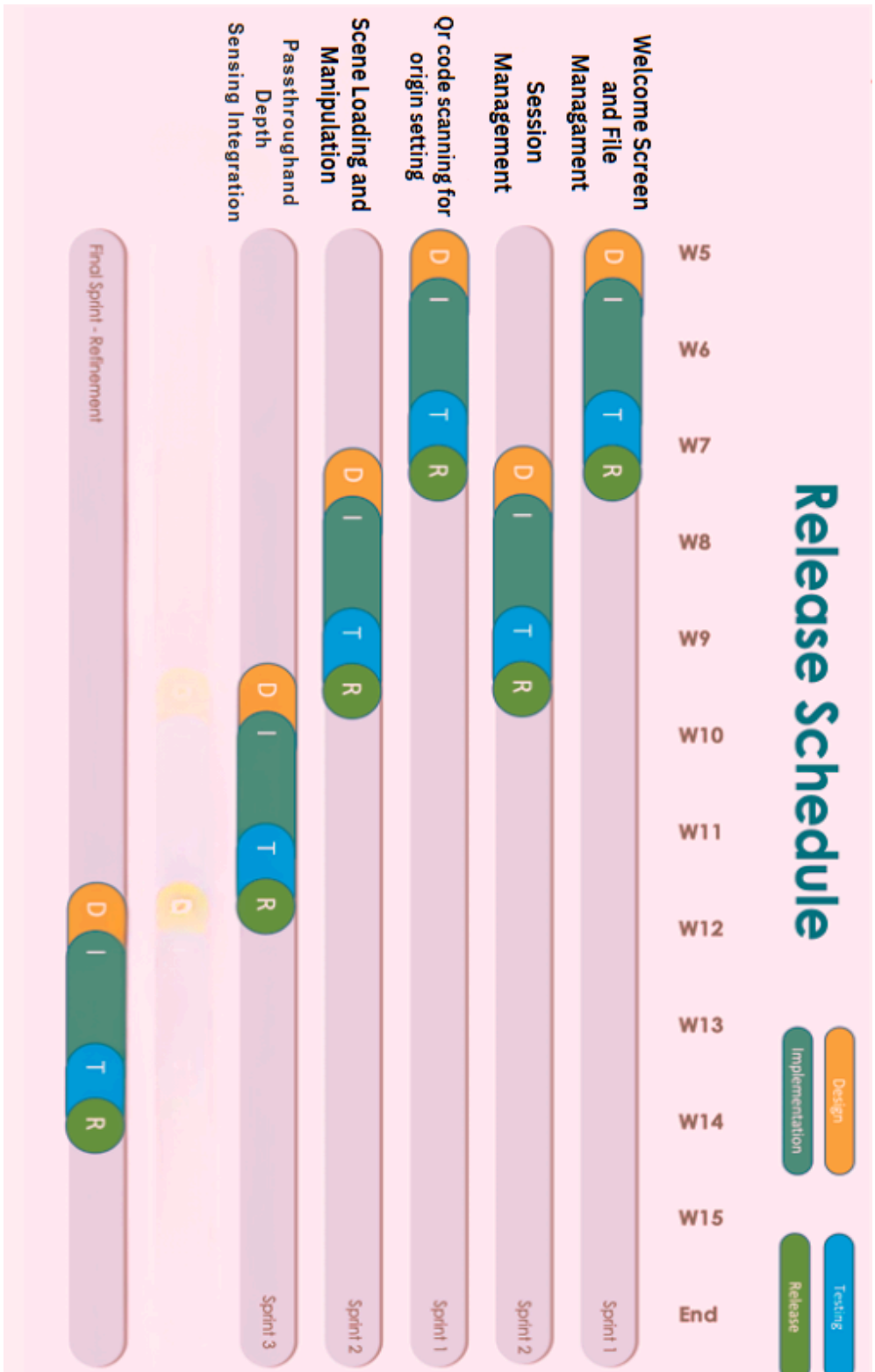
High-Level Business Functions

ID	High Level Business Function	Ranking
BF1	User Authentication and Session Management	Essential
BF2	File Management	Essential
BF3	Scene Interaction and Manipulation	Essential
BF4	Pass through and Depth Sensing Integration	Essential
BF5	Collaboration and Multi-User Interaction	Essential

Functional Requirements

ID	User Story	Priority	Acceptance Criteria
F001	As a user, I want the application to display a welcome screen featuring the ForensicXR logo and then prompt me to select a 3D file from the device, so that I can initiate my investigation session.	Essential	- The welcome screen displays for a set duration before transitioning to file selection. - Only FBX/OBJ files are accepted. - The file selection interface is intuitive.
F002	As a user, I want to choose whether to create a new session or join an existing session via Proton, so that I can collaborate with other investigators.	Essential	- The system offers clear options to create or join a session. - A unique session ID is generated for new sessions. - Session data synchronizes across users.
F003	As a user, I want to scan a QR code/marker to set the origin (0,0,0) of the scene, ensuring that all participants view the scene from their respective location.	Essential	- QR code scanning activates correctly when requested. - The scanned marker accurately sets the session origin. - The origin is shared consistently across all users.
F004	As a user, I want the system to load and spawn the scene from the selected file and allow me to move or resize it, so that I can adjust the scene to my space during a shared session.	Essential	- The LiDAR scene loads correctly from the file. - Users can move or resize the scene via an intuitive interface. - Adjustments updated in real time for all participants.
F005	As a user, I want to use Meta Quest 3's passthrough and depth sensing features while interacting with the scene, so that I can navigate my physical environment safely.	Desirable	- Passthrough mode enables users to view their physical surroundings during the session. - Depth sensing reliably detects obstacles. - Image quality and stability are maintained.

Development Release Schedule



Cost Benefit Analysis for LiDAR Crime Scenes VR Project

Description	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Development Costs	-\$5,800	\$0	\$0	\$0	\$0	\$0
Operations and Maintenance Costs	-\$200	-\$300	-\$800	-\$900	-\$1200	-\$1,400
Cumulative Costs	-\$6,000	-\$6,100	-\$6,600	-\$6,700	-\$7,700	-\$7,200
Benefits from System Operation	\$0	\$1,000	\$2,500	\$6,000	\$10,000	\$15,000
Cumulative Benefits	\$0	\$1,000	\$3,500	\$9,500	\$19,500	\$34,500
Cumulative Cost vs Benefits	-\$6,000	-\$5,100	-\$3,100	\$2,800	\$11,800	\$27,300

Explanatory Notes

Development Costs: Initial one-time expenses including software, licensing (Vuuforia, 8th Wall and Wikitude) fees, hardware devices (Quest 3, PC), and labor (student time at an estimated \$25/hr).

Operations and Maintenance Costs: Continuous expenses such as software updates, 3D model update, Proton/VR/AR/XR SDK and software (Photon, Unity) monthly subscription, and hardware maintenance.

Cumulative Costs: Yearly aggregation of total project expenses.

Benefits from System Operation: Increased productivity for crime scenes users, enhanced training efficiency, reduced field costs, and improved investigative accuracy.

Cumulative Benefits: Total accumulated financial and operational benefits.

Cumulative Cost vs Benefits: Net financial position of the project indicating a positive return starting Year 3.

Alternative Solution - HoloLens 2

This alternative involves continuing the development of the existing holographic crime scene application on the Microsoft HoloLens 2 platform. By leveraging the established code base and prior development work, the project will focus on refining core functionalities such as multi-user session management, scene spawning and interactive scene manipulation. These enhancements are intended to better support forensic investigations. Although the HoloLens 2 is a proven platform, this approach involves optimising the existing application for current operational requirements while making only minimal enhancements to extend its functionality where possible.

Business Case

Continuing the development of the HoloLens 2 capitalises on the previous investments and established user workflow. This option offers a shorter development cycle and lower initial costs because much of the foundational work has already been completed. This may appeal to those who have existing HoloLens 2 hardware and are familiar with its operation. This reduces training and transition costs.

Cost Benefit Analysis

Description	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Development Costs	-\$15,000	\$0	\$0	\$0	\$0	\$0
Maintenance Costs	-\$800	-\$1,000	-\$1,300	-\$1,700	-\$2,100	-\$2,500
Cumulative Costs	-\$15,800	-\$16,800	-\$18,100	-\$19,800	-\$21,900	-\$24,400
Operational Benefits	\$0	\$1,500	\$3,000	\$7,000	\$11,500	\$17,500
Cumulative Benefits	\$0	\$1,500	\$4,500	\$11,500	\$23,000	\$40,500
Net Benefit	-\$15,800	-\$15,300	-\$13,600	-\$8,300	\$1,100	\$16,100

Explanatory Notes

Development Costs: High initial expenses for HoloLens hardware, specialized(XR) SDKs, also software, licensing and skilled labor (approx. \$28/hr) due to unsupported and complex technology.

Maintenance Costs: Elevated ongoing expenses from manual updates, third-party support, and challenges due to lack of official Microsoft support. software (Photon, Unity) monthly subscription

Cumulative Costs: Rising expenses highlight financial risks of unsupported technology.

Operational Benefits: Productivity and cost reductions from advanced VR capabilities, tempered by higher complexity.

Cumulative Benefits: Benefits remain significant, assuming effective management of technical challenges.

Net Benefit: Breakeven delayed to Year 4, illustrating risks and trade-offs with unsupported platforms.

Risk and Constraint

Platform Discontinuation: It is important to note that while the HoloLens 2 approach provides an immediate and cost-effective solution, the long-term viability is limited by the discontinuation of support for the HoloLens 2 platform. This is a significant risk that could affect maintenance and future gesture enhancements.

Short-Term Suitability: This solution is best suited as a short-term solution for organisations seeking rapid deployment and cost saving. They should recognise that future technological updates or platform transition may be necessary as support for the HoloLens 2 ceases.

Technology Transition Costs: Relying on a platform approaching its end-of-life introduces the risk of incurring additional costs and resource investments when transitioning to a new platform in the future. Stakeholders need to consider these potential expenses in their long-term planning.

Conclusions

The recommended solution is to transition to the Meta Quest 3 platform. This approach overcomes the limitations of the outdated Microsoft HoloLens 2 by harnessing the advanced passthrough and depth sensing capabilities of the Meta Quest 3. This not only greatly improves user experience and scene engagement, but also meets the evolving needs of modern forensic investigations.

By developing the project in Unity and leveraging the support of our industry partners, The Mobile Apps Man and ForensicXR, we ensure that the solution is scalable, cost-effective and innovative. The accompanying cost-benefit analysis demonstrates strong financial sustainability, with a positive return projected from Year 3 as a result of improved operational effectiveness and reduced research expenses. This calculated investment in state-of-the-art virtual technology will help forensic experts enhance the precision and collaboration in their investigations.

The LiDAR Crime Scenes for VR project on the Meta Quest 3 platform offers significant corporate value and technological innovation while embodying a forward-thinking strategy. It addresses both current operational requirements and future scalability, making it the preferred solution despite the attractive short-term benefits of continuing with the HoloLens 2.

Reference

Project Documents

Western Sydney University 2025, COMP3018 Professional Experience – Assessment Guide, Western Sydney University, viewed 21 March 2025, available upon request.

Western Sydney University 2025, Project Brief – PA2520: LiDAR Crime Scenes for VR, Western Sydney University, viewed 21 March 2025, available upon request.